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INFO324 - Length Of ICU Stay Prediction, Report

Introduction & Motivation

Problem Definition

The project predicts how long a patient will stay in the ICU using data gathered from the MIMIC-III Clinical database. Instead of predicting the exact number of hours, we aim to determine patients who stay longer than 7 days versus those who don't.

Motivation

Many hospitals today are often overcrowded and underfunded, with too few ICU beds available for the number of patients who need them. Because of this, clinicians have to make quick decisions about how to allocate limited space and resources. A model that predicts ICU length of stay can help identify which patients are likely to require more intensive, longer-term care and which patients might be treated and discharged more quickly. By recognizing and planning for longer-term patients, hospitals would be able to allocate their resources more effectively to improve overall efficiency.

Project Goal

The goal of this project is to build models that are both accurate and interpretable using demographic, clinical, and treatment features taken from MIMIC-III. Medical decisions need to be made swiftly and effectively, identifying a patient that might need a longer ICU stay before it happens ensures that the patient receives the care they need to prevent a longer stay than necessary.

Data Wrangling & Feature Selection

Datasets Used from MIMIC-III

All data is derived from the MIMIC-III 1.4 dataset. The list below describes the variables used to build patient-level features. The dataset is a list of the admitted patients to the I CU of Beth Israel Deaconess Medical Center in Boston. This data is composed of data from 2001 and 2012.

Data Set Exploration:

- **ADMISSIONS:** Every unique hospitalization for each patient in the database (defines **HADM_ID**)
- **ICUSTAYS:** Every unique ICU stay in the database (defines **ICUSTAY_ID**)
- **PATIENTS:** Every unique patient in the database (defines **SUBJECT_ID**)

- **MICROBIOLOGYEVENTS**: Microbiology measurements and sensitivities from the hospital database
 - **INPUTEVENTS_MV**: Intake for patients monitored using the iMDSof Metavision system while in the ICU
 - **INPUTEVENTS_CV**: Intake for patients monitored using the Philips CareVue system while in the ICU
 - **LABEVENTS**: Laboratory measurements for patients both within the hospital and in outpatient clinics
 - **DIAGNOSES_ICD**: Hospital assigned diagnoses, coded using the International Statistical Classification of Diseases and Related Health Problems (ICD) system
 - **PRESCRIPTIONS**: Medications ordered, and not necessarily administered, for a given patient
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- age (from PATIENTS.csv via DOB and ICUSTAYS.csv INTIME): captures baseline frailty and physiologic reserve, which strongly influences LOS.
 - gender (from PATIENTS.csv): accounts for sex of patient. treatment, reactions to certain treatments, and outcomes.
 - ethnicity (from ADMISSIONS.csv if available, otherwise set to "UNKNOWN"): proxies social, genetic, and access-to-care factors that may affect LOS.
 - admission_type (from ADMISSIONS.csv): distinguishes elective vs emergency/urgent admissions, which have very different expected length of stay.
 - first_careunit (from ICUSTAYS.csv): identifies the ICU type (e.g., medical vs surgical), helps find more serious reasons for stay which could prolong length of stay (surgical may stay longer than medical).
 - micro_S_count and micro_R_count (from MICROBIOLOGYEVENTS.csv INTERPRETATION): summarize susceptibility and resistance patterns, which can influence difficulty of treating infection which would increase overall length of stay
 - vasopressor_amount, total_input_amount, received_vasopressors (from INPUTEVENTS_MV.csv via ITEMID and AMOUNT columns): identify whether the patient receives vasopressors (drugs used during emergency situations) as well as the dosage, capturing treatment intensity and hemodynamic instability that can prolong LOS.
 - any_abnormal_flag, num_abnormal_labs, has_abnormal_labs (from LABEVENTS.csv FLAG): determine whether or not the patient has any notes marked “abnormal” on their lab events chart. Often associated with longer LOS.
 - num_diagnoses, has_critical_diagnosis (from DIAGNOSES_ICD.csv ICD9_CODE): count the number of diagnoses and flag critical diagnoses, reflecting diagnostic complexity and severity of illness that can increase the likelihood of a prolonged ICU stay.

We cleaned the raw data by fixing timestamp issues, removing invalid entries, and calculating each patient's total ICU stay in hours. The `long_stay` label was created from this, and is classified as any total stay for a patient that is longer than 7 days. We also had to ensure that any missing data was taken care of, so for most of the data it was replaced with a value of: "UNKNOWN" to fix holes in the data.

Methodology

We trained supervised models to predict ICU length of stay using a linear regression and a random forest regressor built on features derived from demographics, admissions data, lab/microbiology summaries, prescriptions, diagnoses, and treatment intensity. For the regression task, we randomly split the cohort at the encounter level into 80% training and 20% test using `train_test_split` (`test_size=0.2`, `random_state=42`), and we reduced the size of the larger cohorts (to 8000–15000 iterations) to keep the computation running smooth while also giving an adequate split. We reported mean absolute error, root mean squared error, and R^2 : MAE to find directly interpretable error in days spent, RMSE to highlight larger errors that are more impactful for operations, and R^2 to summarize how much variation in length of stay is explained by variables. Also, to predict a long ICU stay (≥ 7 days), we used the same feature set with logistic regression and random forest classifiers, again with an 80/20 split but this time making it stratified to keep balance, and we reported the same AUROC, F1, precision, and recall scores.

Results & Analysis

RandomForest:

AUROC: 0.9297

F1: 0.6625

Precision: 0.5512

Recall: 0.8300

LogisticRegression:

AUROC: 0.8628

F1: 0.5236

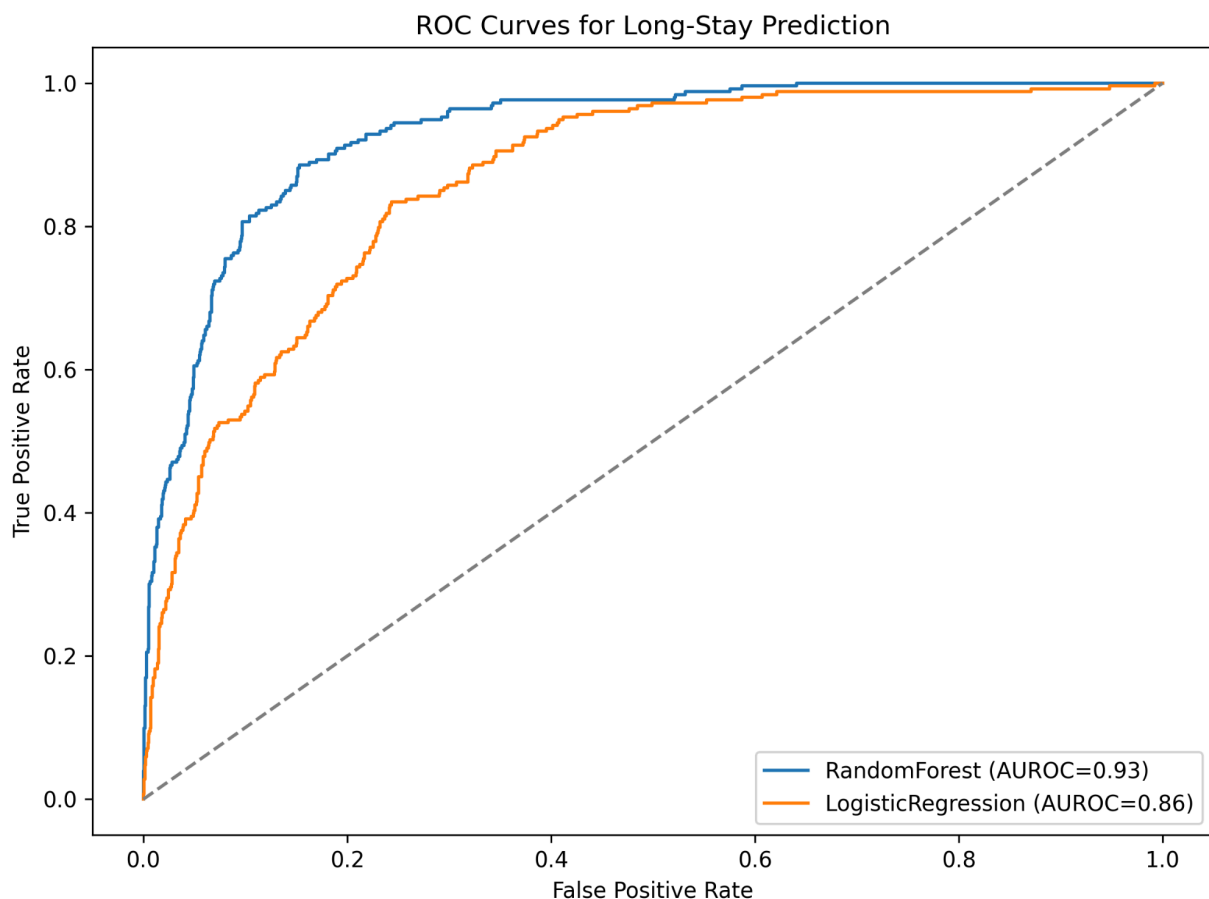
Precision: 0.3816

Recall: 0.8340

Our results show that routinely collected ICU data can support reasonably strong discrimination for predicting long stays (≥ 7 days). The random forest classifier achieved an AUROC of 0.93, compared to the 0.86 for logistic regression, indicating that it more effectively separates patients who will have prolonged ICU stays from those who will not cross a range of decision thresholds. This suggests that the interactions between admission characteristics, infection markers, and treatment-intensity features carry meaningful signals that are not fully captured by the linear model, which may be why it is not as effective as the random forest model.

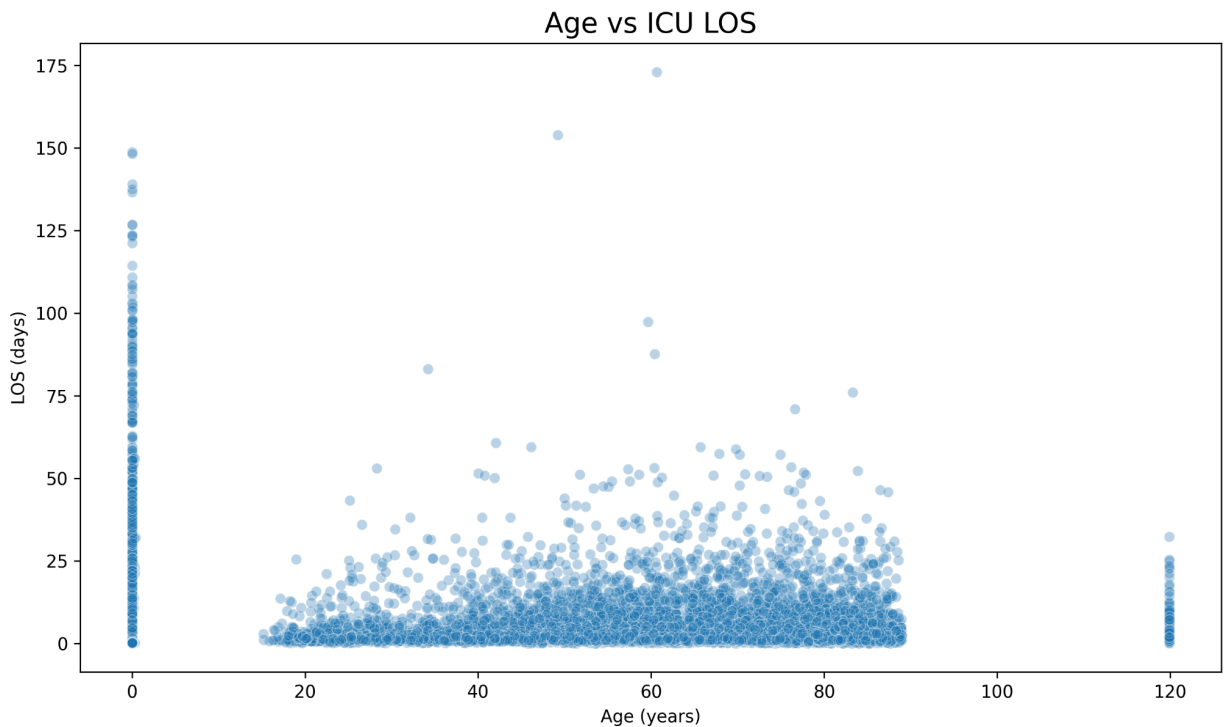
The random forest also reached higher overall F1 (0.66 vs 0.52) and precision (0.55 vs 0.38) while maintaining high recall (0.83 vs 0.83), meaning that when it flags a patient as likely to experience a long ICU stay, that alert is more likely to be correct than with logistic regression, without sacrificing much sensitivity. For clinicians and bed-managers, this translates into fewer “false alarms” and a more manageable set of high-risk patients to prioritize for early discharge planning, social work involvement, or closer monitoring of complications.

The stronger performance of the random forest, however, suggests that a combination of these factors are predictive of prolonged stays, than a linear comparison. Overall, the results suggest that tree-based models are promising for operational forecasting of ICU length of stay, but any model should be used as a decision-support tool rather than a standalone determinant of care.

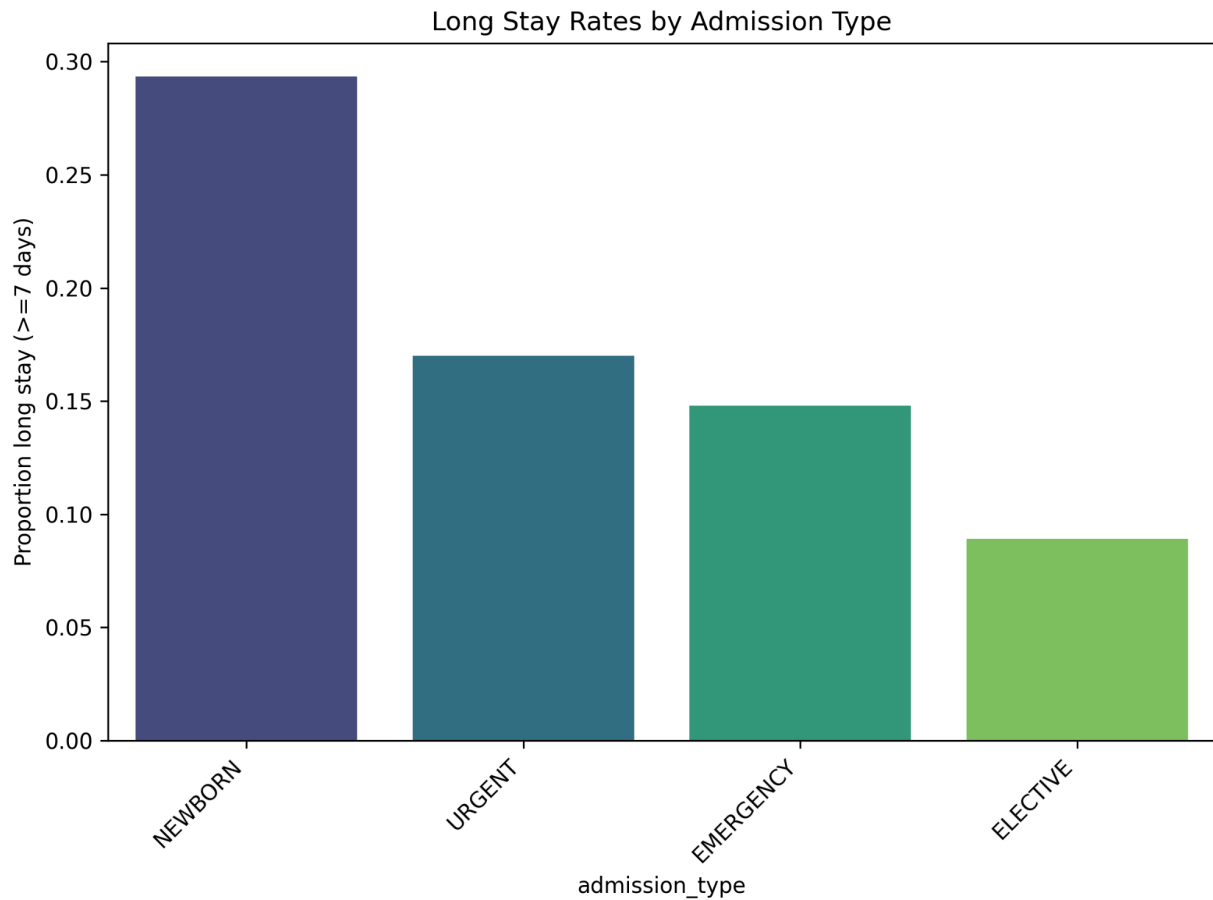


This figure compares how well the random forest and logistic regression models discriminate between ICU stays that are long (≥ 7 days) and those that are not. The ROC curves plot the true positive rate (correct, positive assessments) against the false positive (incorrect, positive assessments) rate across all possible decision thresholds. AUROC curves that bow further toward the top-left corner indicate better discrimination. The random forest model outperforms the logistic regression model and achieves a higher AUROC (0.93 vs 0.86), meaning it is better at determining high-risk patients. The accompanying metrics at the chosen 0.5 threshold show that the random forest also has higher F1 (0.66 vs 0.52) and precision

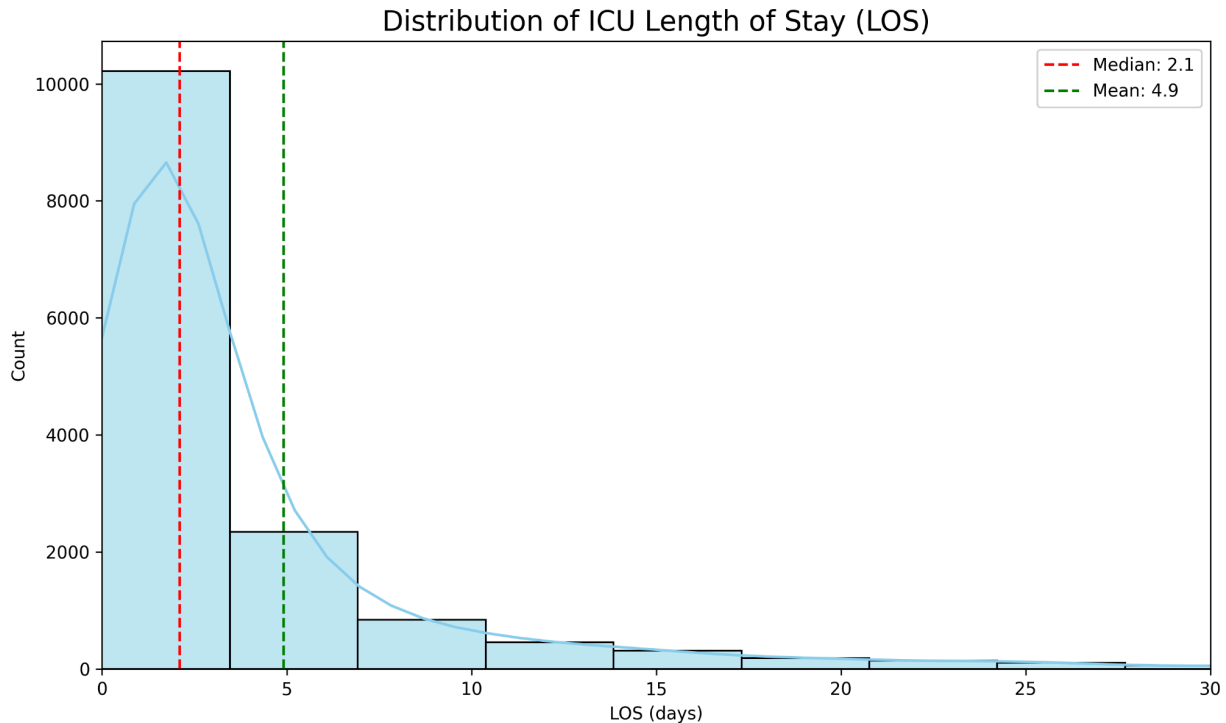
(0.55 vs 0.38) while maintaining similar recall (~ 0.83), so it identifies a comparable proportion of true long-stay patients but with fewer false positive reports. The random forest model is the more useful tool for flagging a manageable subset of patients who are genuinely at high risk of prolonged ICU occupancy.



The age versus ICU length of stay (LOS) scatter plot shows the relationship between patient age and the number of days spent in the ICU for each admission. Most points are clustered between ages 20–90 with LOS under about 30 days, indicating that for the majority of adult patients, LOS is relatively short and does not increase linearly with age. The outliers in the data are made intentionally in the dataset to protect the identity of the patients. Overall, the plot suggests that while very young and very old patients can have prolonged ICU stays, age alone does not explain a significant amount of LOS variability, so many other factors are important in understanding what influences length of ICU stay.



This bar chart shows the proportion of ICU admissions that result in a long stay (≥ 7 days) for each admission type. The y-axis represents the fraction of patients within that admission category who meet the long-stay definition. Newborn admissions have the highest long-stay rate, followed by urgent and emergency admissions, while elective admissions have the lowest probability of a prolonged ICU course. This could highlight that newborn and urgent patients should be prioritized for early discharge planning to ensure that the length of stay is as efficient as possible.



The histogram of ICU length of stay (LOS) shows how many ICU admissions fall into each LOS bin, with the y-axis “Count” representing the number of patients whose LOS falls within that range of days. The distribution is highly right-skewed: most ICU stays are short, with a median of about 2.1 days (red dashed line), while the mean is higher at around 4.9 days (green dashed line) because a relatively small number of patients have very prolonged stays that pull the average to the right. For clinicians and operations teams, this plot highlights that short stays are far more common than long ones, but a long tail of complex patients consumes a disproportionate amount of ICU bed-days, which is exactly the part of the distribution our prediction models aim to identify early.

Ethics & Critique

Since the MIMIC-III dataset comes from a single hospital, which limits how the model can be used to apply to other patients from other hospitals. Since different hospitals can use entirely different systems, this model is fitted specifically for this data and could not be applied to any other system. Also, since the situation of every patient is different, there is no model that can accurately portray the experience of a patient in the ICU. There are so many factors that influence the length of an ICU stay that it is difficult to determine what features are useful in a predictor model. Some features (like treatment intensity) may be relatively subjective. Also, any data point in the set is prone to human error from the hospital staff that submitted them initially. Further, this project is somewhat limited by the data we are using. There is very limited data to be gained from one hospital, especially data collected over 10 years ago. With how much innovation there is in technology and medicine, it is possible that the average length of stay could decrease if monitoring took place today.

Future Work

Ideally, this work would be expanded to include many more features that would have an effect on the length of an ICU stay. For example, the surgical history of the patient is not considered in the current model. This is likely to affect the overall performance of the model because someone with an extensive surgical history would be more likely to have a longer stay in the ICU due to more significant concerns than those without. We would also expand this work to include data from other hospitals since the dataset is somewhat limited in its collection based on location and amount of data collected. One thing that could be interesting is to include vital signs but I was constrained by both time and the processing power of my computer to go through all of the CHARTEVENTS.csv file.

Reflection

One of the most important skills we learned while doing this project was learning how to properly clean a data set. With the missing values in some of the data sets there were often errors when trying to access the values that we wanted to. After properly cleaning the data and figuring out what to look for in different files it became easier to manage. We will definitely take this skill and use it in the future, it lets us get a better understanding of the data that we are using. This project also gave us a better understanding of troubleshooting specific errors. We experienced a lot of traceback errors with the code that we had not seen before that mostly stemmed from use of the data set. This project has actually helped a lot with problem solving skills in informatics and we will be using the data management skills learned here in the future.